Characterization for Structure-Property Understanding: From Molecular Mechanisms to Polymer Materials

GM R&D Research Seminar | May 21, 2026

Speaker: Chuan Pin (Pin) Chen

Postdoctoral Appointee, Argonne National Laboratory

From Mechanism to Materials

2018 – 2024

Graduate Teaching/Research Assistant
Ph. D. in Chemistry

Mechanistic characterization:
Synthesis, NMR, isotope, Raman

May 2024 – Aug 2024

Intern Chemist

Optical characterization:
Dye synthesis, UV–Vis, FL

2025 – Present

Postdoctoral Appointee

Polymer characterization:
Synthesis, fabrication, EIS
Laboratory setup

10+	13	1
years research	publications	patent pending

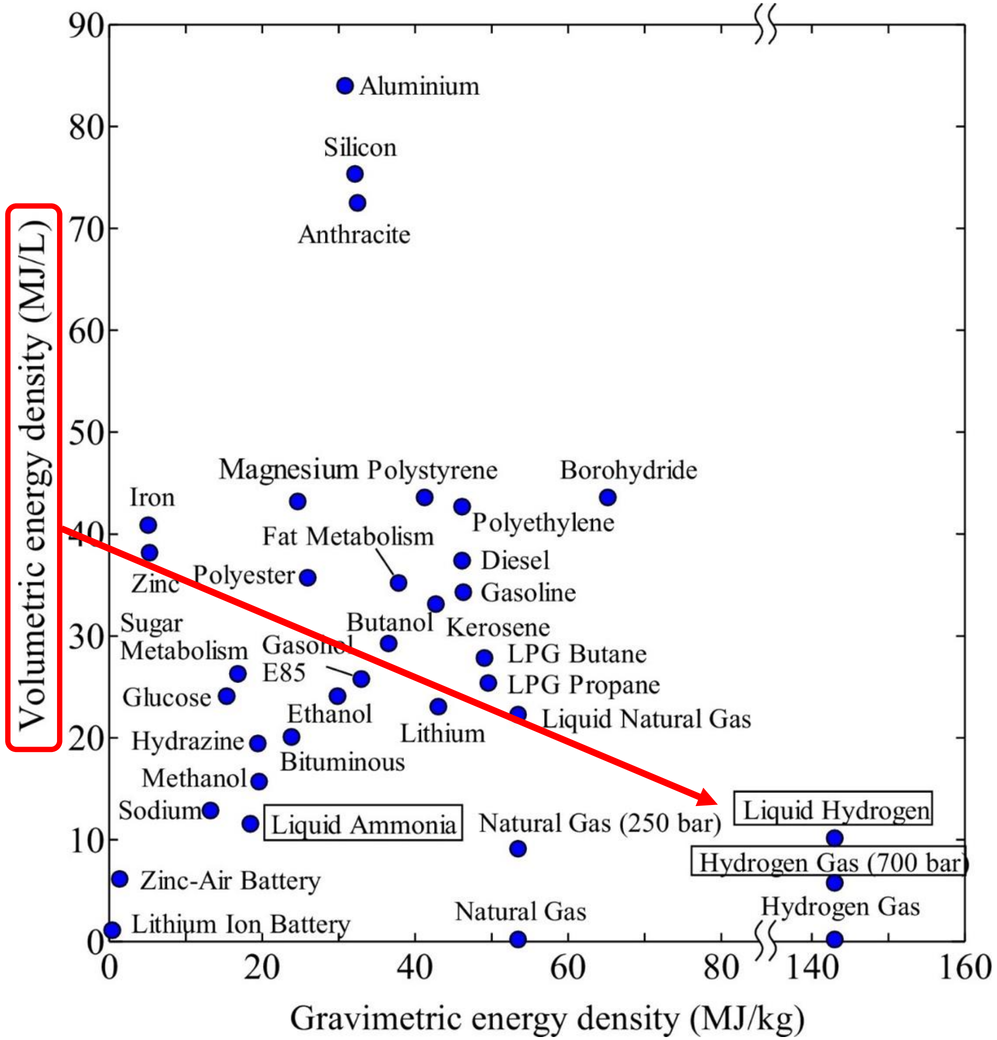
Why Study Ammonia Oxidation?

- Hydrogen (H_2) is a promising clean energy carrier

$$2\,H_2O \xrightarrow{\text{Electricity for Electrolysis}} O_2 + 2\,H_2$$

Solar

Wind



- **Challenge:** difficult to store and transport
- **Solution:** store H_2 in chemical bonds → Ammonia (NH_3)

Economic Times, 2019; Scientific American, 2019; Vox, 2019; *Proc. Combust. Inst.* 2019, 37, 109–133.

NH_3 as a Storage Medium

- Easy to liquefy, distribute

$$\frac{3}{2}\,H_2 \xrightarrow[\text{H}_2\text{ Transportation/Storage}]{} NH_3$$

- Reverse reaction, ammonia oxidation

$$\frac{1}{2}\,N_2 + \frac{3}{2}\,H_2 \underset{}{\overset{\substack{\text{Fe catalyst}\\ 360\text{-}510°\text{C}\\ 150\text{-}175\text{ bar}}}{\rightleftharpoons}} NH_3$$

$$NH_3 \overset{\text{Electrocatalysis}}{\rightleftharpoons} \frac{1}{2}\,N_2 + \frac{3}{2}\,H_2$$

Haber Bosch

Thermal cracking

Efficiency

Continuous working

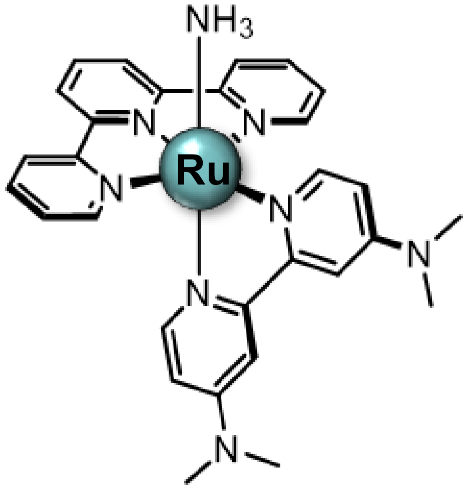
On-demand, small-scale, mild condition, direct ammonia fuel cell (DAFC) applications

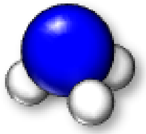
- Energy barrier

(A) *Proc. Natl. Acad. Sci. U.S.A.* **2019**, *116*, 2849–2853. **(B)** *ACS Catal*. **2019**, *9*, 10101–10108. **(C)** *Nat. Chem.* **2019**, *11*, 702–709.

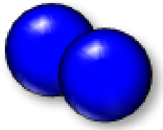
From Catalyst to Characterization Challenge

- First homogeneous electrocatalyst (**$[Ru^{II}(NH_3)]^{2+}$ precatalyst**)

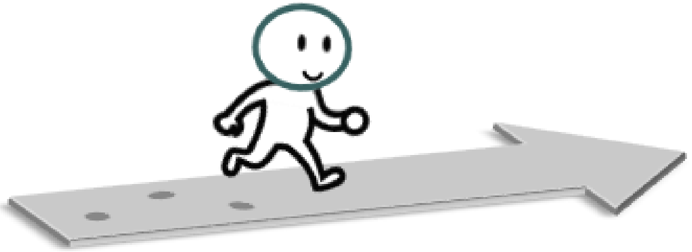




NH_3 $\xrightarrow[\text{room temperature, ambient pressure}]{[Ru^{II}(NH_3)]^{2+}}$ **N_2**



- Mechanism → guides future catalyst design



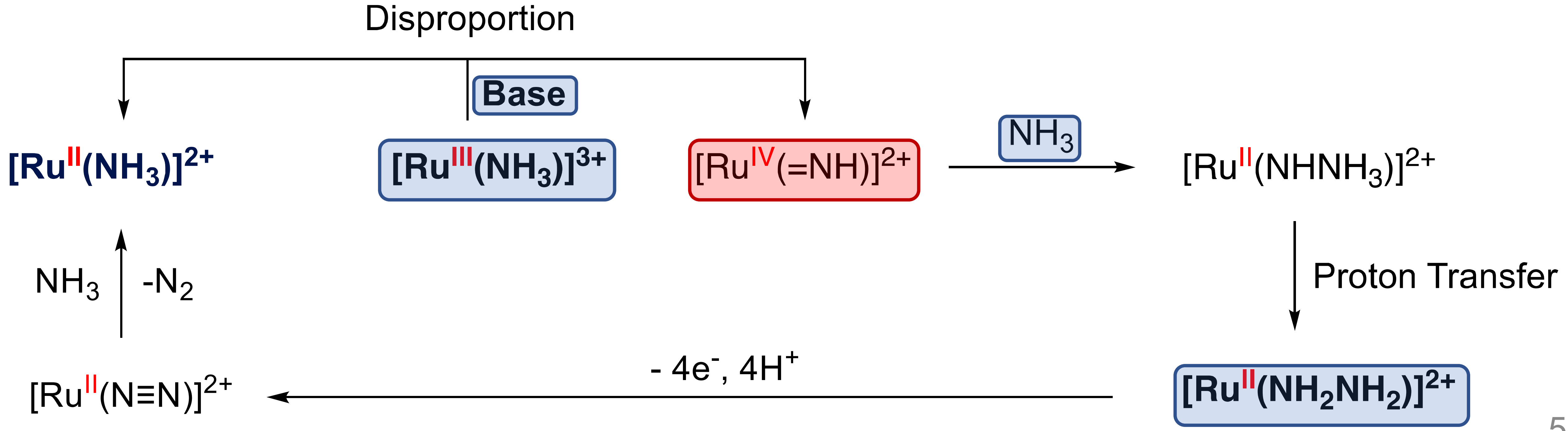
- **Characterization problem:** how do we identify **what is hard to track?**

J. Am. Chem. Soc., ***1981***, *103*, 5579–5581; *Proc. Natl. Acad. Sci. U.S.A.* **2019**, *116*, 2849–2853; *Inorg. Chem.* **2025**, *64*, 13599–13603

Initial Proposed Mechanism Based on NMR

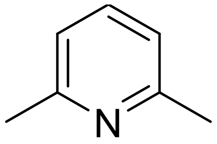
- **Ru^{2+} to Ru^{3+}** is the first electrochemical step.
- NMR Results (Not balanced)

$$[Ru^{III}(NH_3)]^{3+} \xrightarrow[\text{DCM-}d_2,\ -80^\circ C]{\text{excess } NH_3} [Ru^{II}(NH_3)]^{2+} + [Ru^{II}(NH_2NH_2)]^{2+} + \textbf{Unknown}$$



Proc. Natl. Acad. Sci. U.S.A. **2019**, *116*, 2849–2853.

Initial Proposed Mechanism Based on NMR

- Use innocent base instead of NH_3: Collidine  → hinder/weak

$[Ru^{III}(NH_3)]^{3+}$ $\xrightarrow[\text{DCM-}d_2,\ -80^\circ C]{\text{excess } NH_3}$ $[Ru^{II}(NH_3)]^{2+}$ + $[Ru^{II}(NH_2NH_2)]^{2+}$ + **Unknown**

Disproportion

Base

$[Ru^{II}(NH_3)]^{2+}$ ← $[Ru^{III}(NH_3)]^{3+}$ → $[Ru^{IV}(=NH)]^{2+}$

$[Ru^{IV}(=NH)]^{2+} \xrightarrow{NH_3} [Ru^{II}(NHNH_3)]^{2+}$

$[Ru^{II}(NHNH_3)]^{2+}$ $\xrightarrow{\text{Proton Transfer}}$ $[Ru^{II}(NH_2NH_2)]^{2+}$

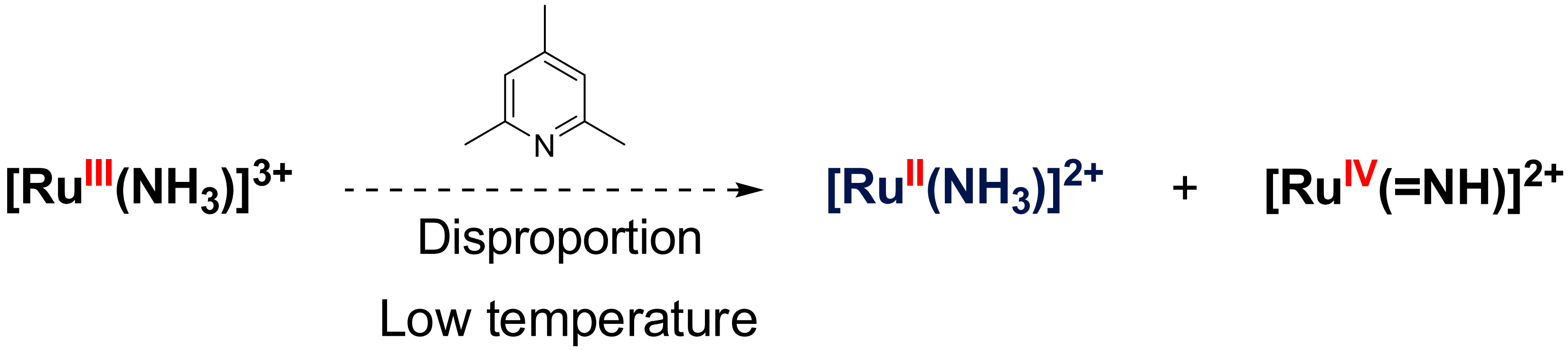
$[Ru^{II}(NH_2NH_2)]^{2+} \xrightarrow{-\ 4e^-,\ 4H^+} [Ru^{II}(N\equiv N)]^{2+}$

$[Ru^{II}(N\equiv N)]^{2+} \xrightarrow{NH_3,\ -N_2} [Ru^{II}(NH_3)]^{2+}$

Proc. Natl. Acad. Sci. U.S.A. **2019**, *116*, 2849–2853.

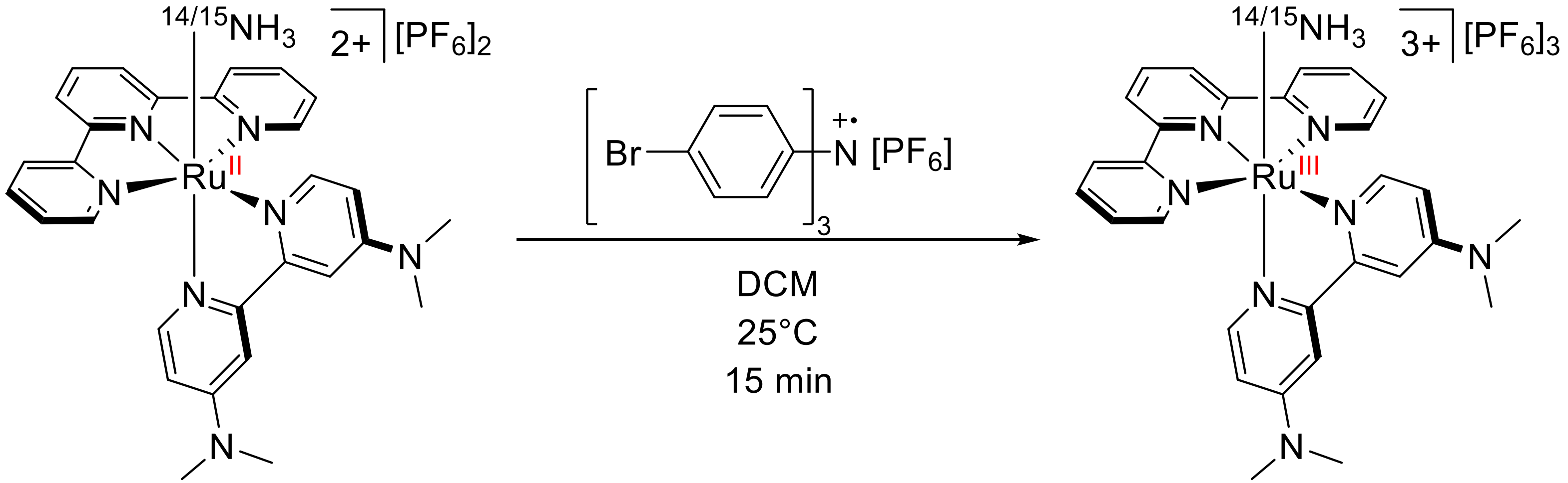
Strategy: Trap, Label, and Identify

- Unstable $[Ru^{IV}(=NH)]^{2+}$



$^{15}NH_3$ labeled

→ fingerprint

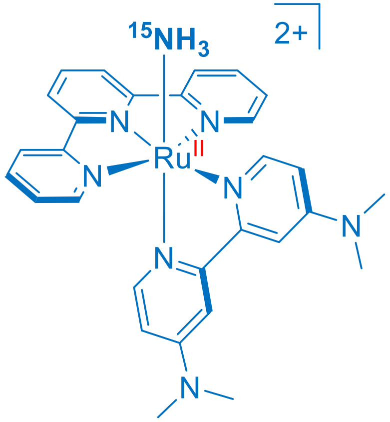


$^{14}NH_3$ 93%

$^{15}NH_3$ 96%

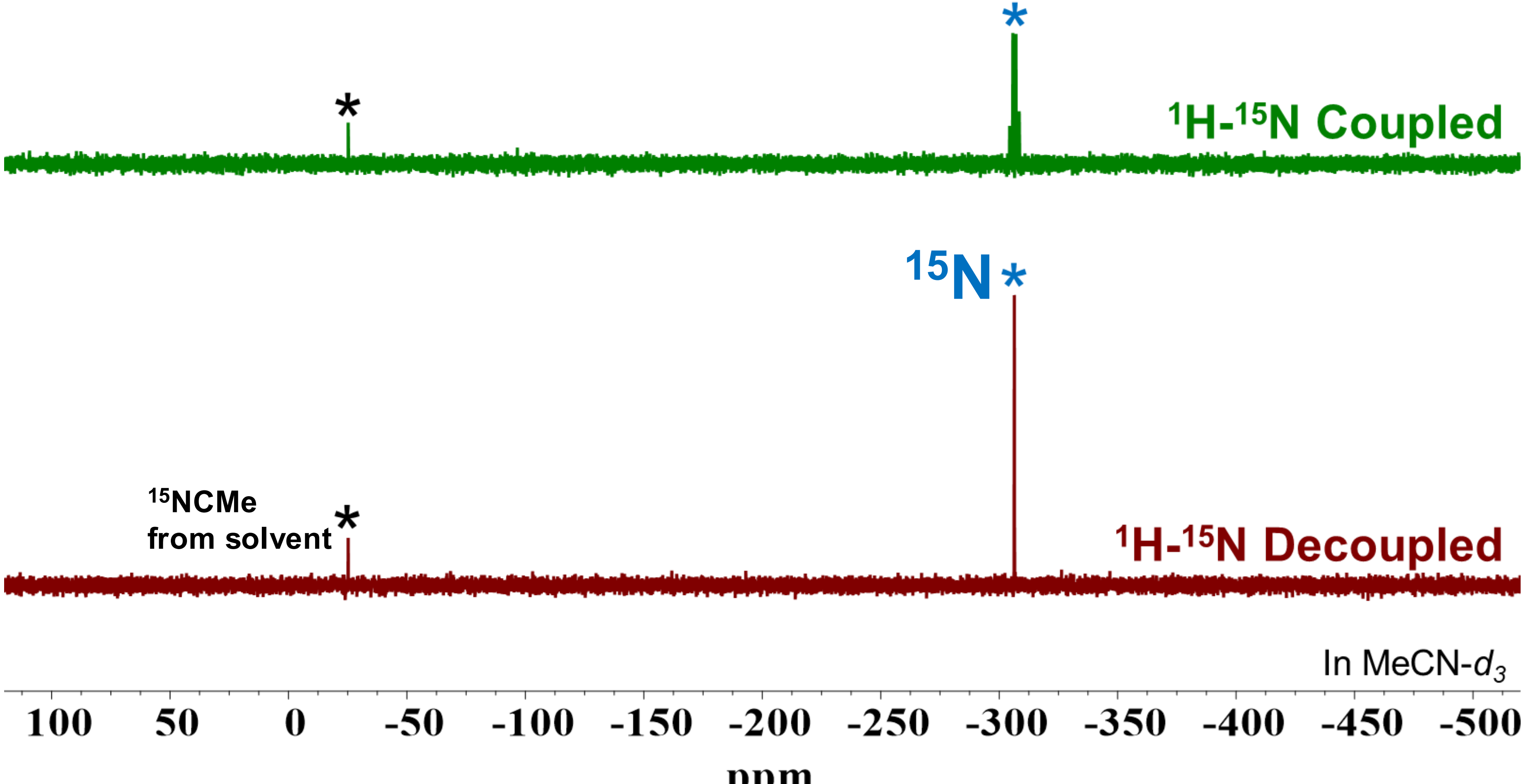
Proc. Natl. Acad. Sci. U.S.A. **2019**, *116*, 2849–2853.

^{15}N NMR as a Fingerprinting Tool



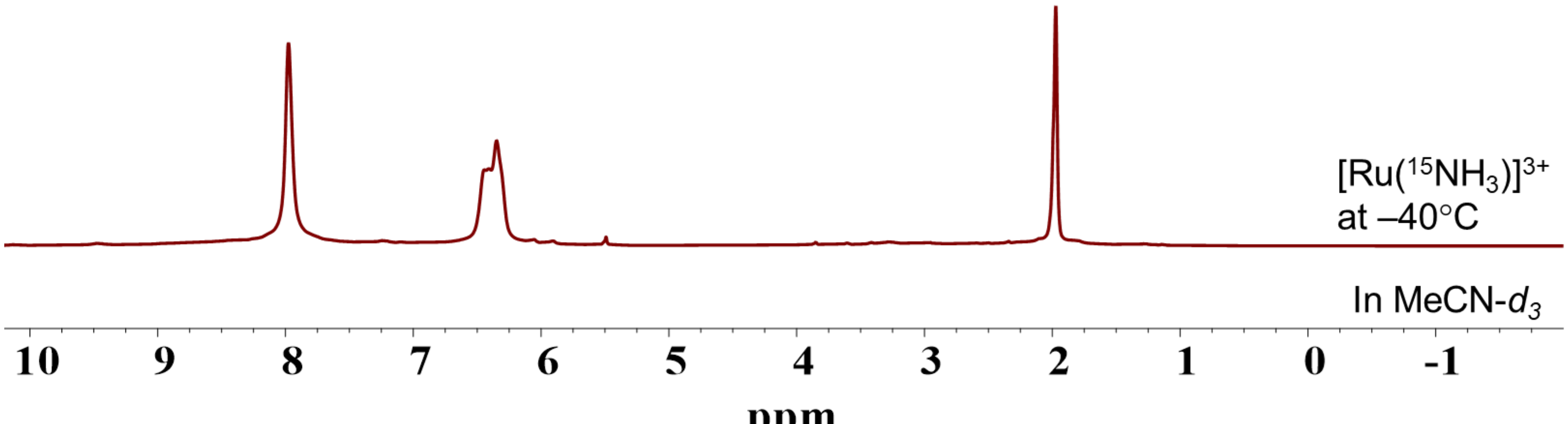
$[Ru^{II}(^{15}NH_3)]^{2+}$

Quartet by coupled with proton
n+1 rule



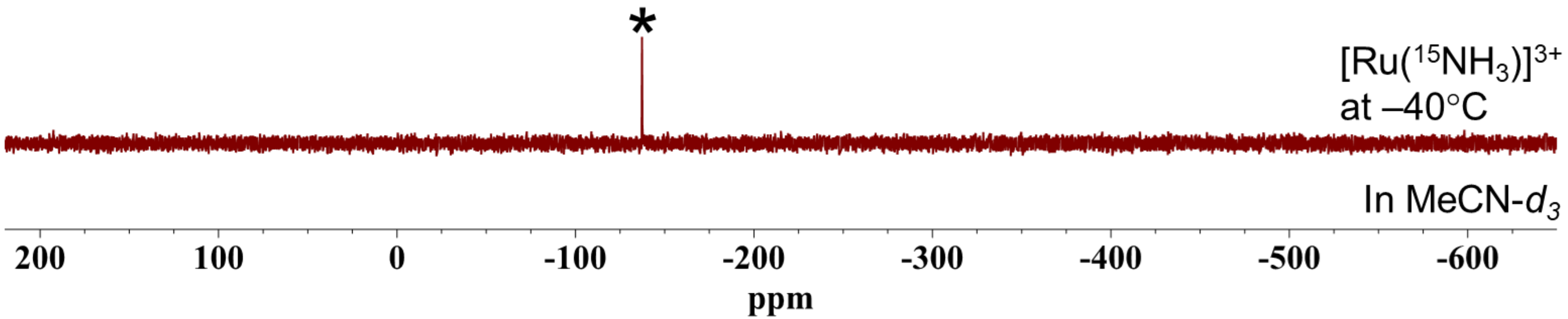
Intermediate Detected: ^1H NMR Evidence at –40°C

- $[Ru(^{15}NH_3)]^{2+}$
- $[Ru(NCMe\text{-}d_3)]^{2+}$
- Interme



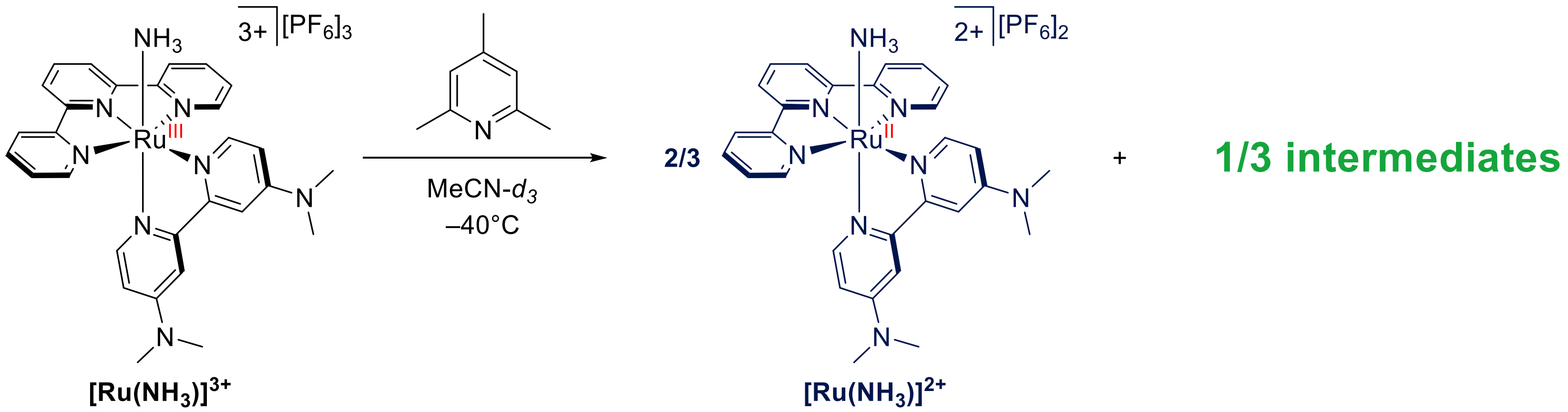
Intermediate Confirmed: ^{15}N NMR Fingerprinting

- Interm



NMR Results and Open Questions

- Based on the $^{1}H/^{15}N$ NMR results.

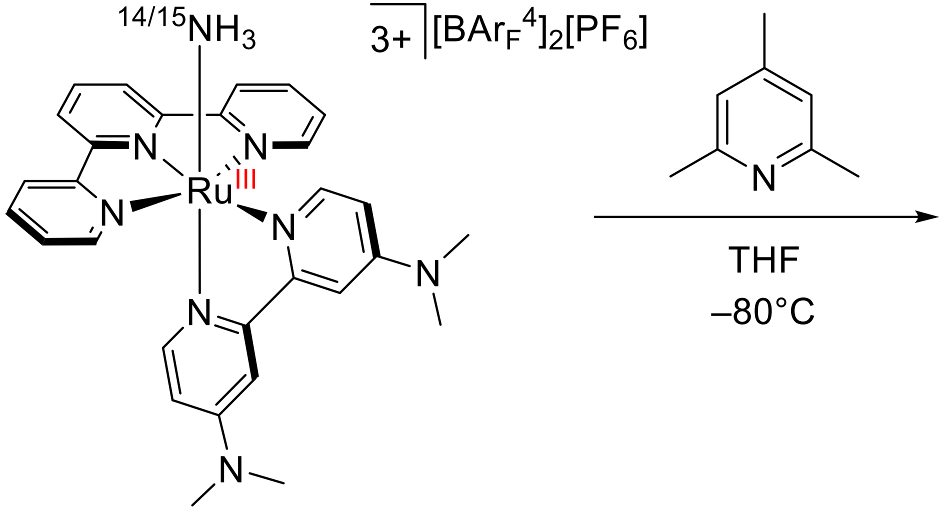


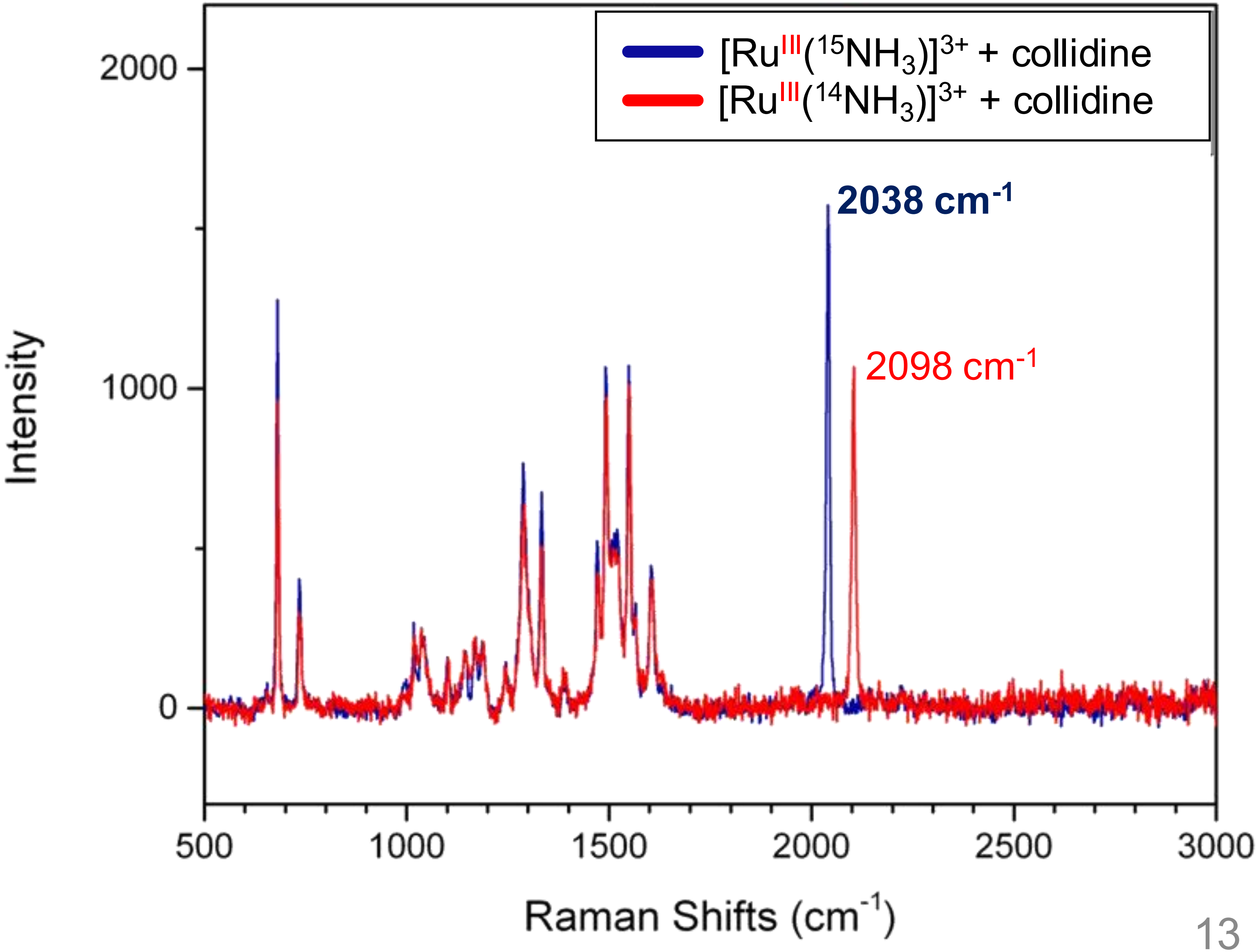
1. What is the structure of **Intermediate**?
2. How does it form? Is it related to $[Ru^{IV}(=NH)]^{2+}$?

Systematic Approach: Narrowing Down the Intermediate Structure

Structure	^{15}N Signals Predicted	IR	Raman
$Ru{-}^{15}N_{\alpha}{\equiv}^{15}N_{\beta}$	2 peaks	Active	Inactive
$Ru{-}^{15}N_{\alpha}{\equiv}^{15}N{-}Ru$	1 peak	Inactive	Active
$Ru{\equiv}^{15}N_{\alpha}$	1 peak	Active	Inactive
$Ru \leftarrow \begin{matrix} ^{15}N\alpha \\ \vert\vert\vert \\ ^{15}N \end{matrix}$	1 peak	Active	Inactive

Raman Confirms: A Dinitrogen-Bridged Complex

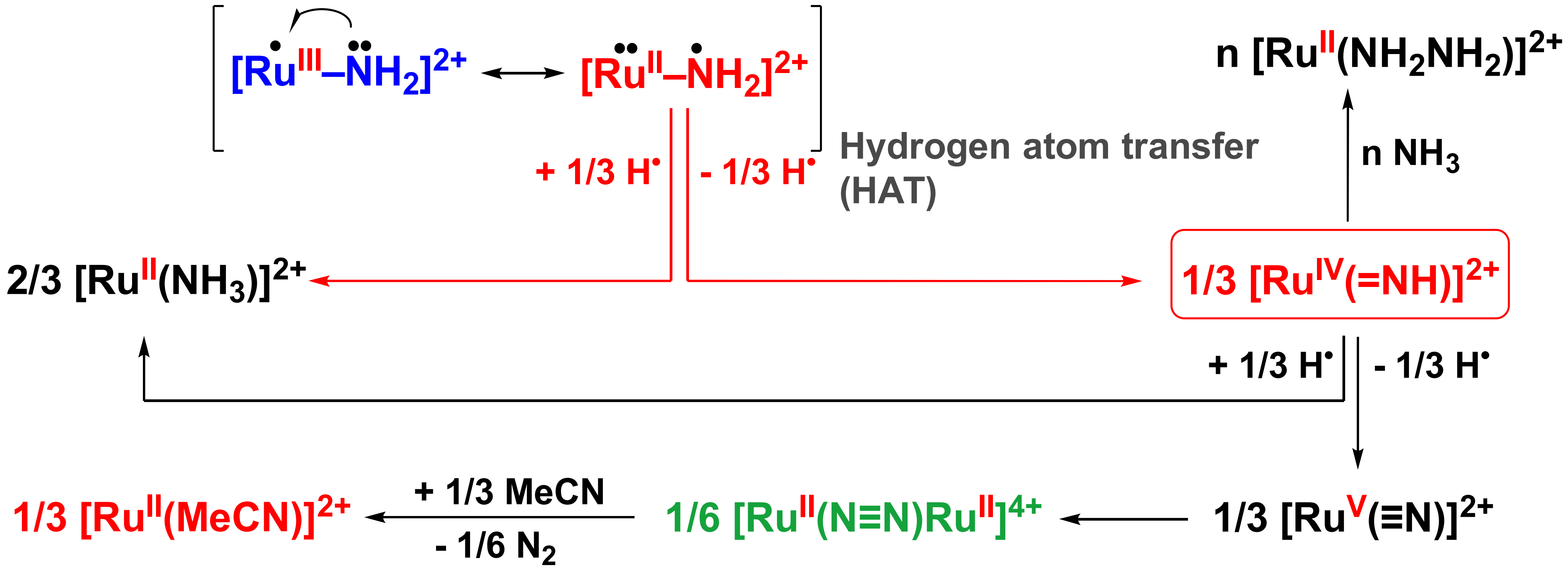




- Free N_2: 2359.6 cm^{-1}
- Expected $^{15}N_2$ on Ru complex: 2027 cm^{-1} (harmonic oscillator)
- **Nitrogen bridged complex**

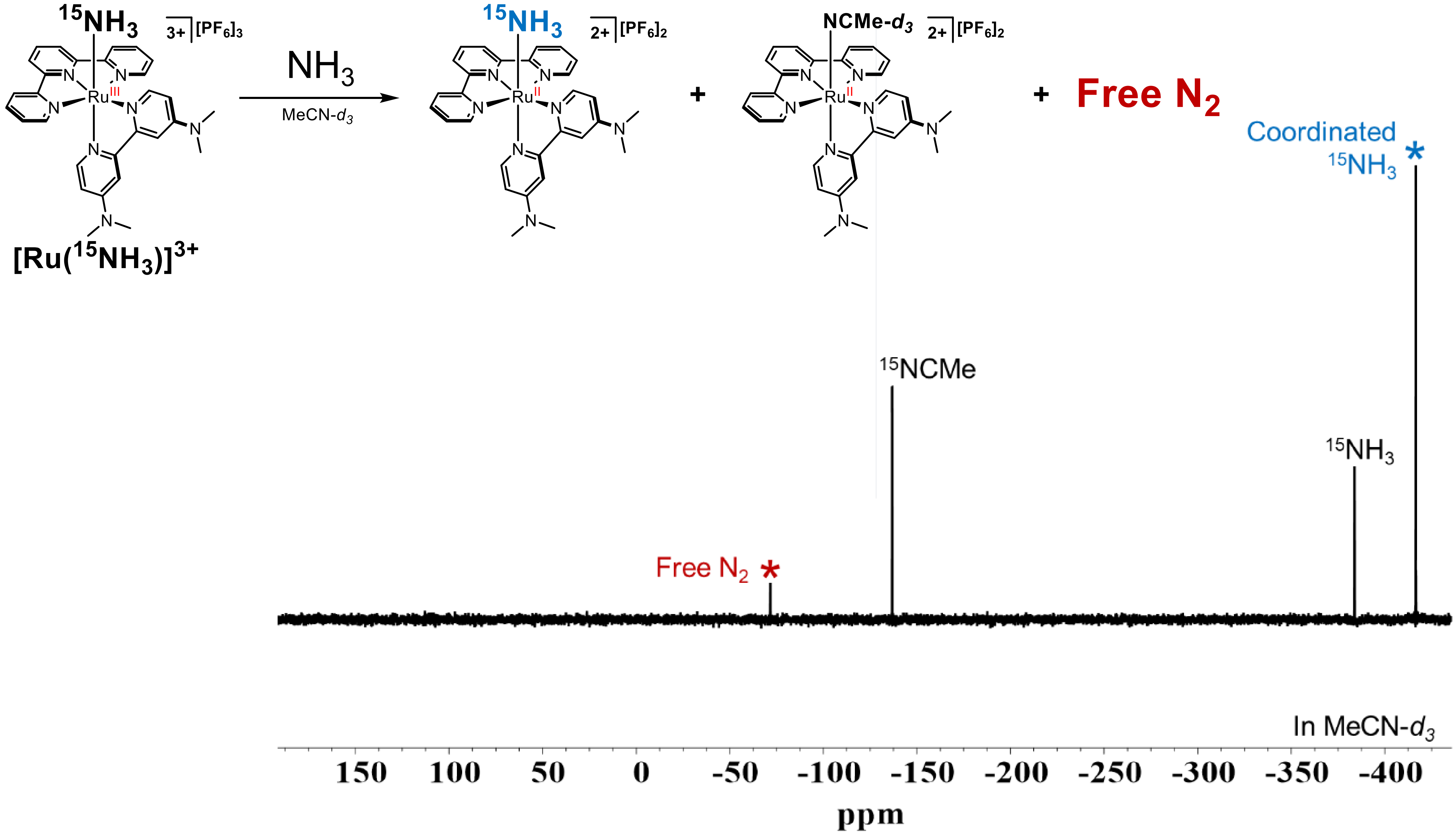
Two Possible Pathways Identified by NMR and Raman

- To explain possible N-N bond formation

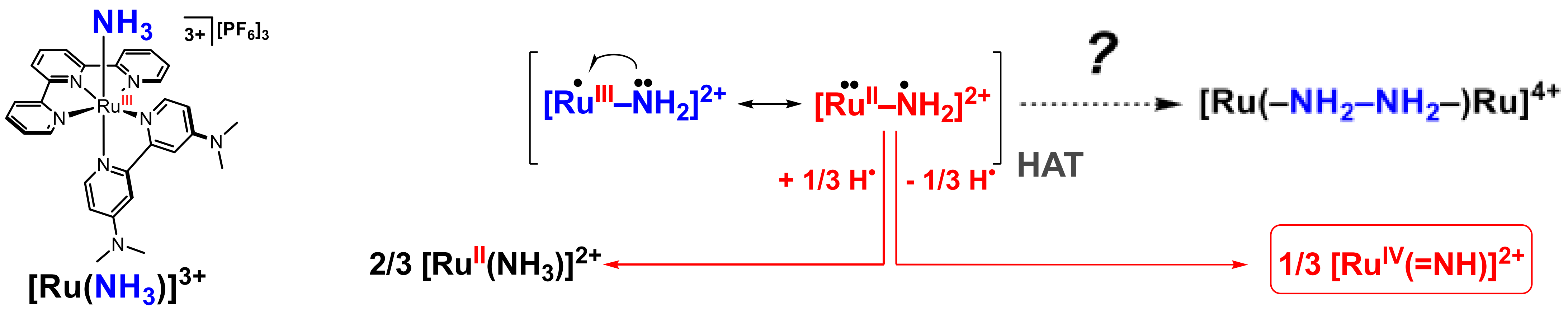


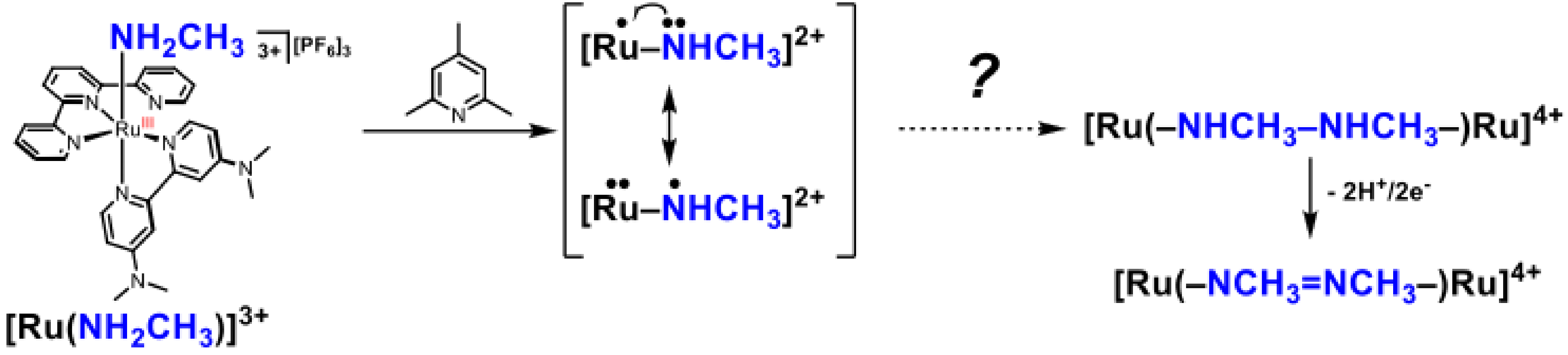
- $[Ru^{III}(^{15}NH_3)]^{3+} + NH_3$
- $\boxed{[Ru^{II}(^{15}NH_2NH_2)]^{2+}}$ and $\boxed{[Ru^{II}(^{15}N{\equiv}^{15}N)Ru^{II}]^{4+}}$

Both Pathways Confirmed: Isotope Labeling Reveals Two N-N Bond Formation Routes



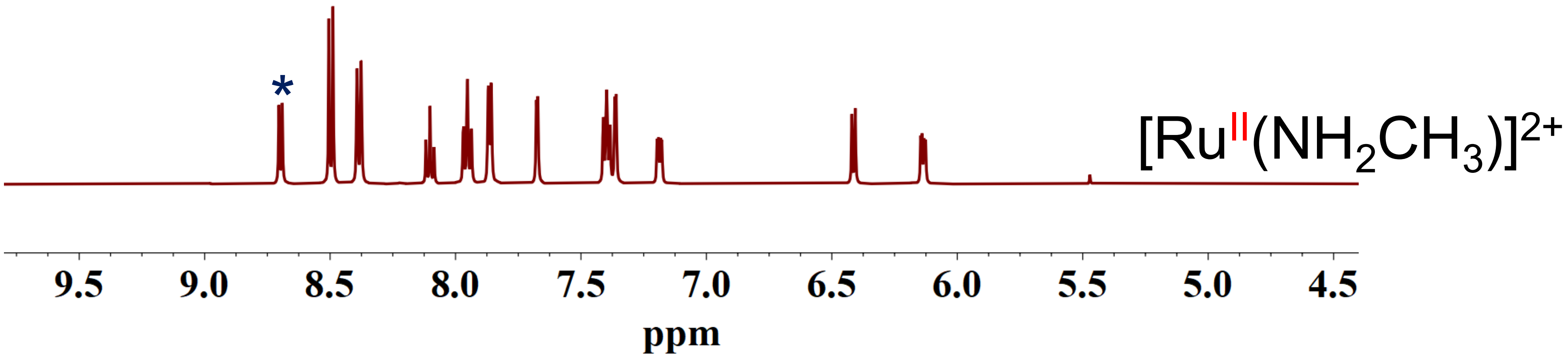
A Third Possible Pathway: Could Radical Homocoupling Also Occur?

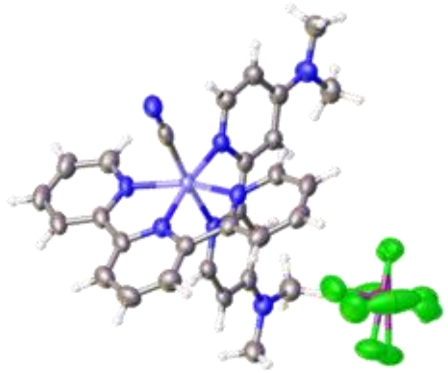




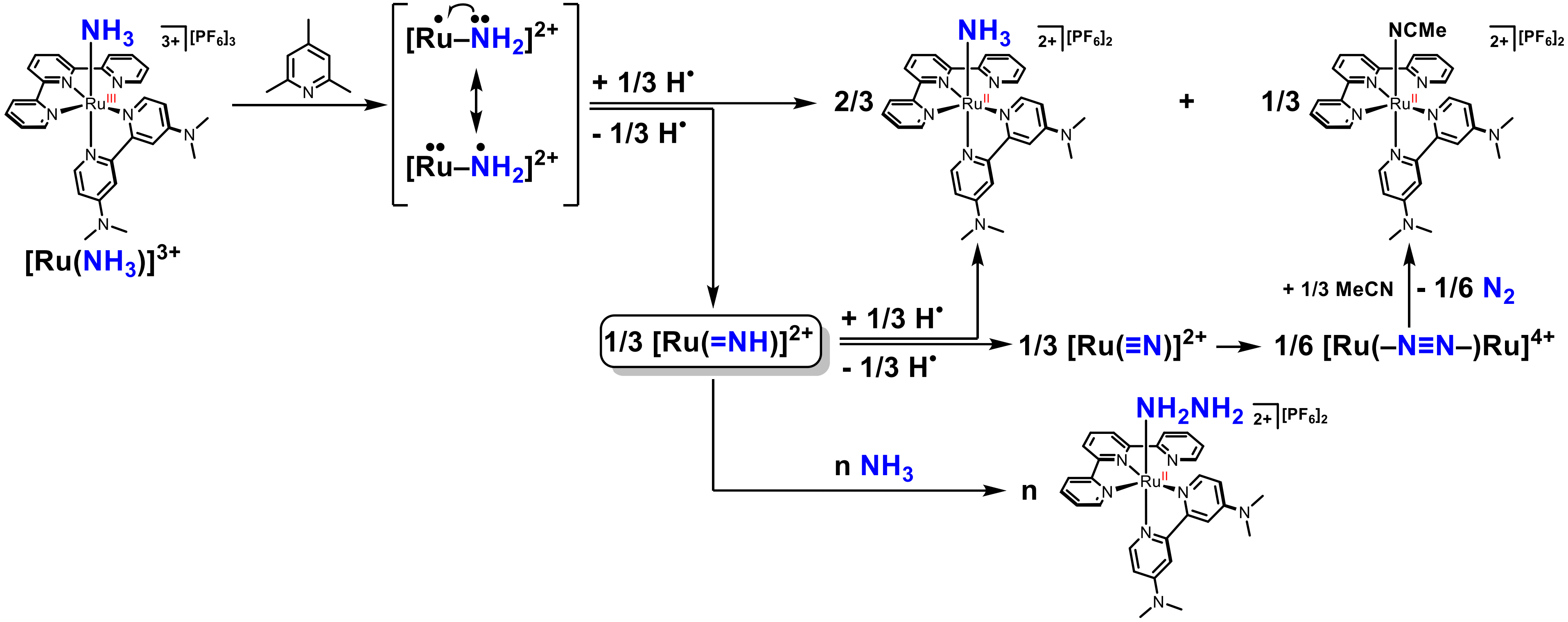
Molecular Design as a Mechanistic Probe: Eliminating the Radical Homocoupling Pathway

- **No [R**





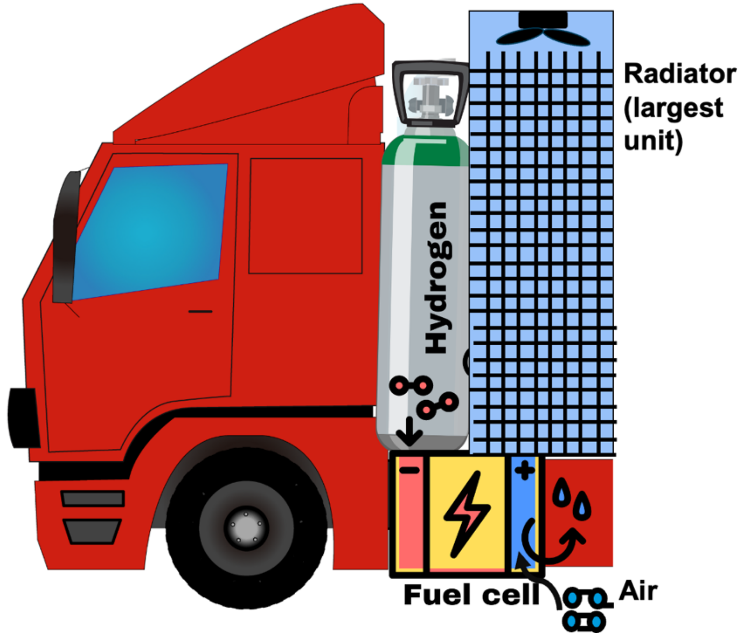
From Characterization to Insight: What This Means Beyond Ammonia Chemistry



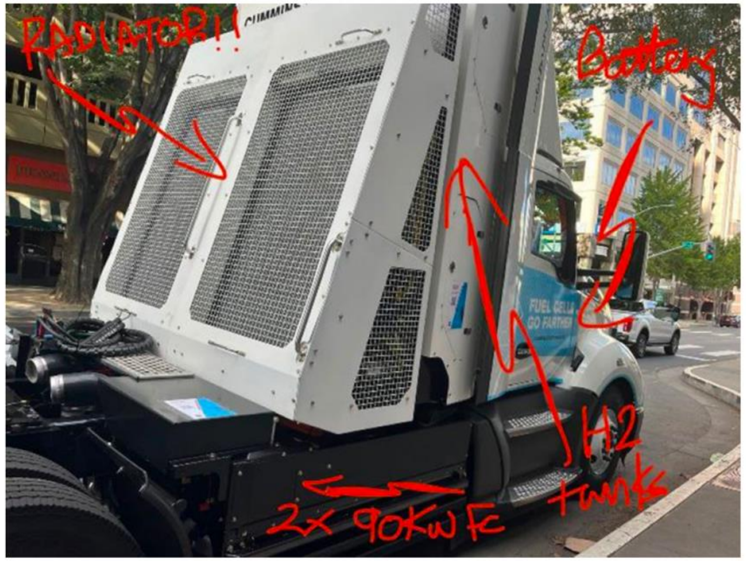
- No single technique was enough ➜ multimodal approach
- *"When you have eliminated the impossible, whatever remains, however improbable, must be the truth." — Sherlock Holmes*

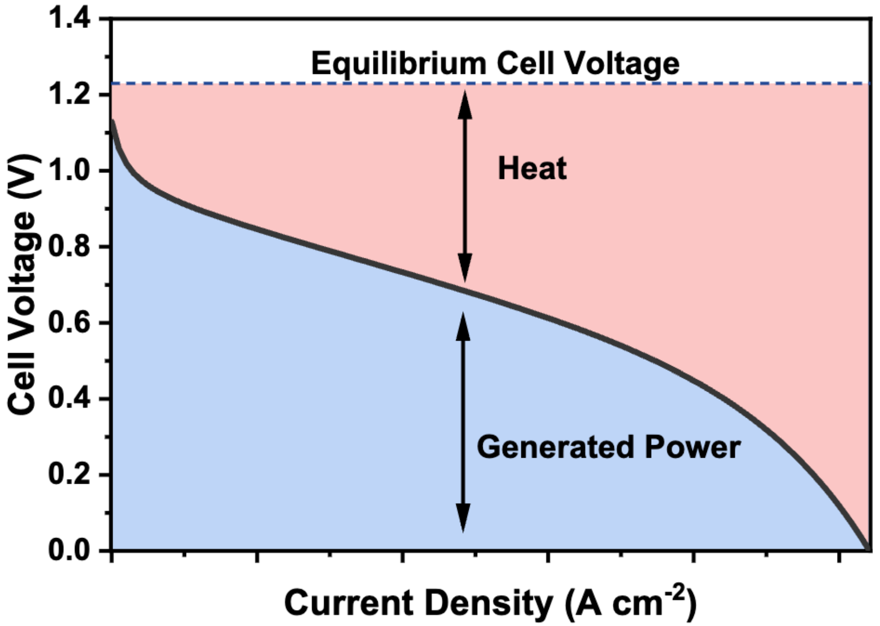
J. Am. Chem. Soc. **2023**, *145*, 26339-26349.

Why Develop Membranes at High Temp/Dry Conditions?



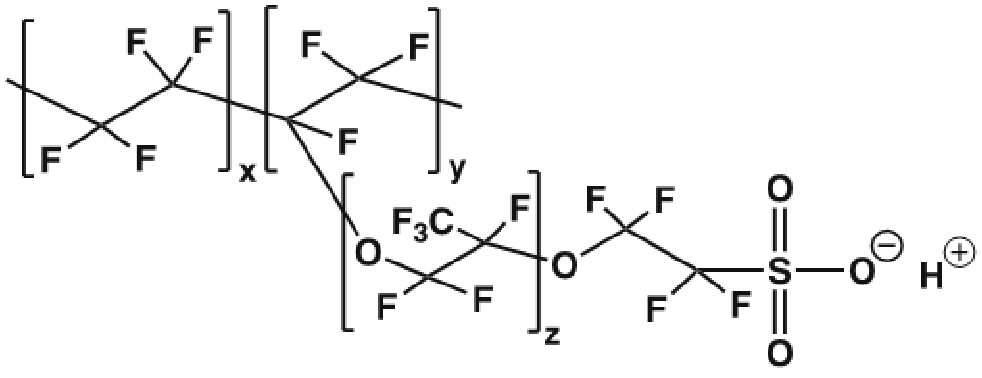
$$\dot{Q} = U \cdot A(T_{stack} - T_o)$$



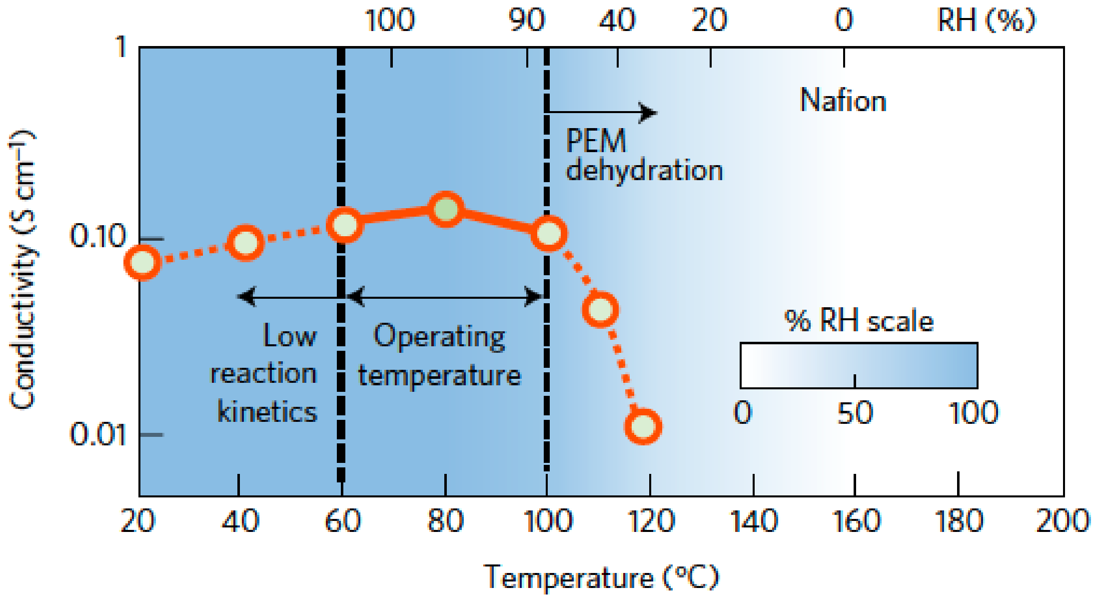


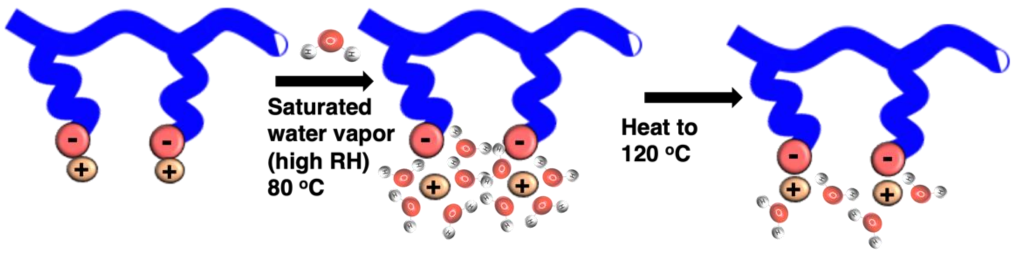
- DOE's LT-PEMFC Target for heat rejection (Q/ΔT) is < 1.45 kW °C^{-1} at 40 °C ambient.
- LT-PEMFCs need to operate > 90 °C to satisfy this requirement when operating at 0.75 W cm^{-2}.
- Hotter operation can enhance electrode kinetics and impurity tolerance

Current Membranes Fail Under Hot and Dry Conditions



Nafion™



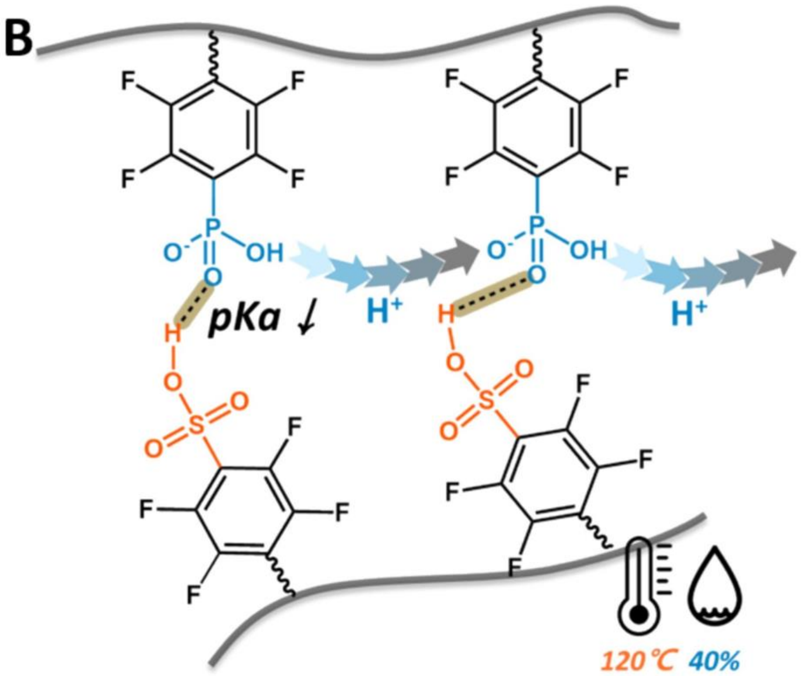


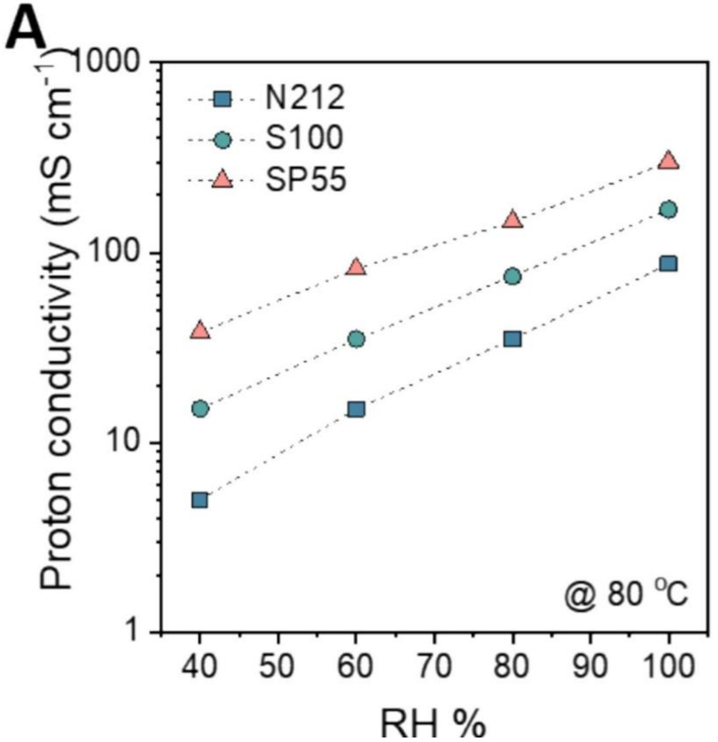
- Excellent proton conductivity **when well hydrated**
- Proton transport **strongly depends on water**
- Membranes lose conductivity

E. De Castro, *L'Innovator Program*; U.S. Department of Energy HFTO Annual Merit Review, 2024.

Combining Two Acid Groups to Engineer Hot/Dry Performance

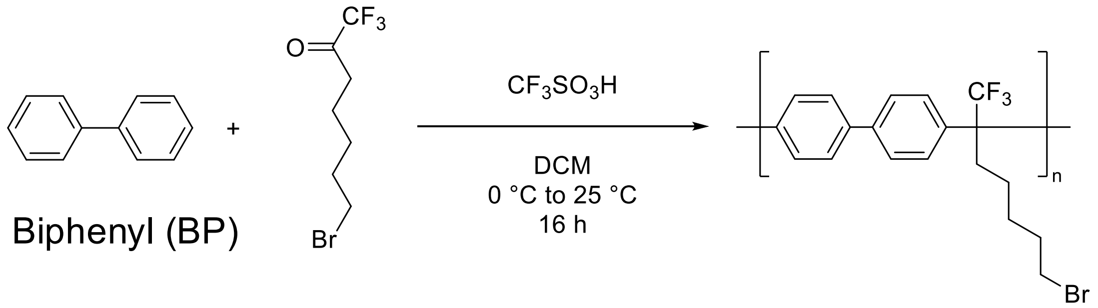
- **Sulfonic acid:** strong acidity, efficient proton dissociation
- **Phosphonic acid:** conducts protons without free water
- Combination should improve conductivity under hot, dry conditions





Chem. Eng. J. **2025**, *522*, 167336

Controlling Molecular Weight: A Critical Parameter for Membrane Fabrication

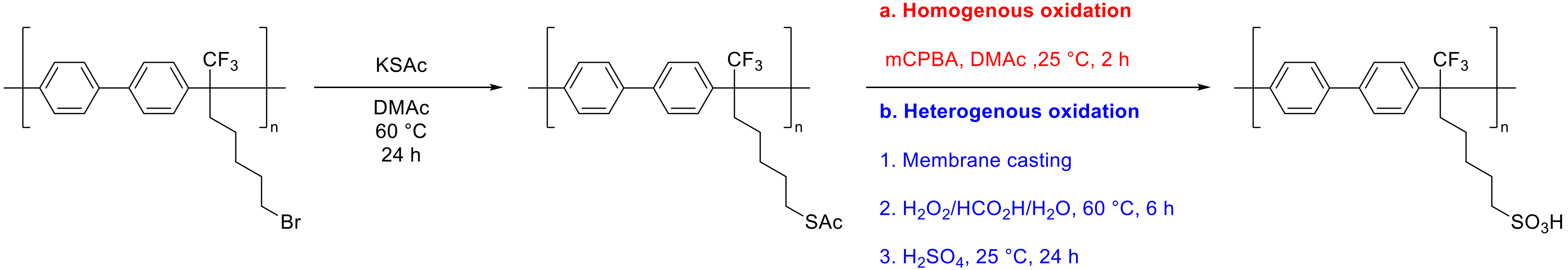


Biphenyl (BP)

BPBr

- Molecular weight control was **"critical"** to balance solubility, membrane formation, and mechanical integrity for CEM polymers.
- **GPC:** Mw 49 kDa; PDI = 1.8

Processing Challenge: Heterogeneous Oxidation Enables Reproducible Membrane Fabrication



BPBr (49 kDa) BPTA BPSA

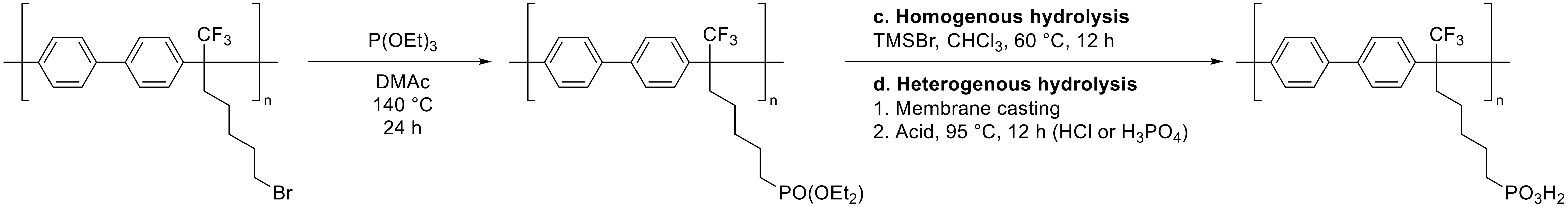
- **Heterogeneous oxidation** – consistent membrane fabrication

10 wt% in DMAc
Centrifuge

BPTA membrane

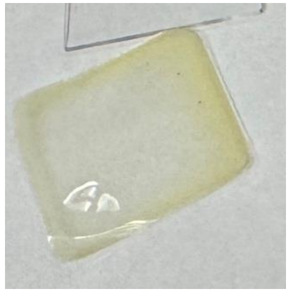
Proton conductivity: 84.5 mS cm^{-1}
(25 °C, water)
IEC (ion exchange capacity): 2.3 mmol g^{-1}
Patent Application

Processing Challenge: Phosphonic Acid Groups Cause Phase Separation

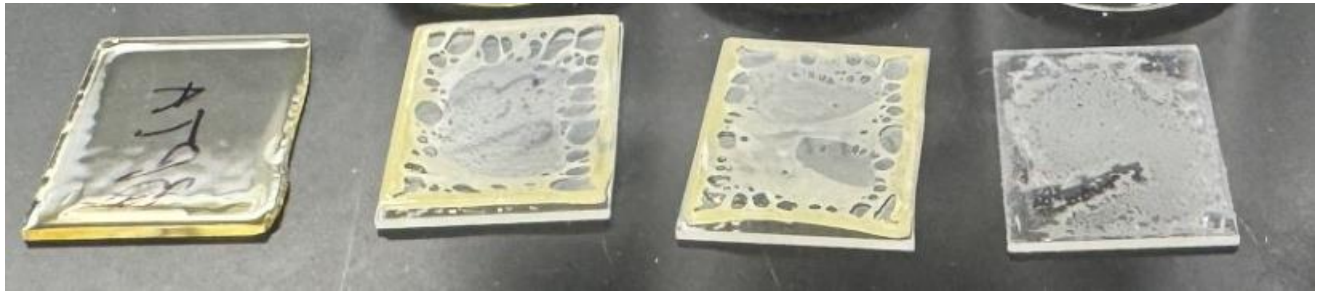


BPBr (49 kDa) BPPO BPPA

- Severe phase separation caused by BPPA blends



[BPTA/BPPO = 7:3]

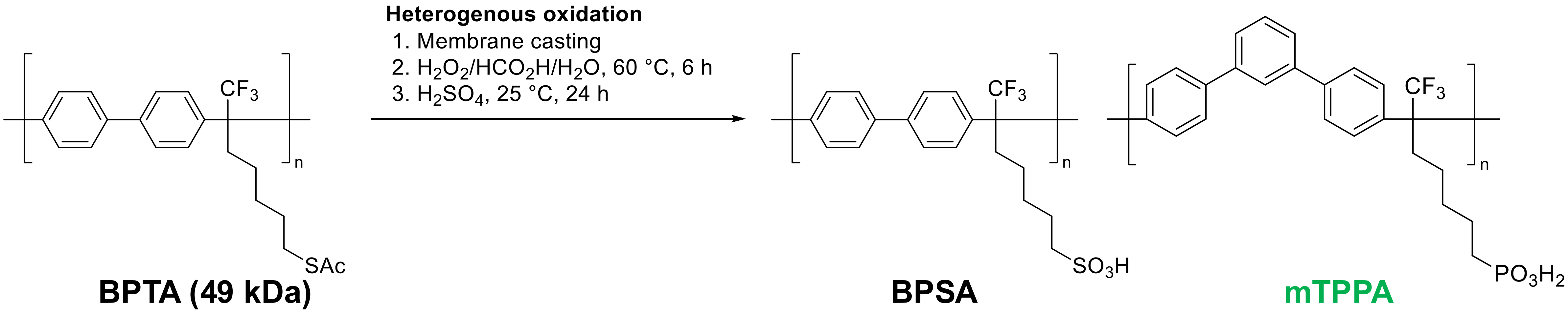


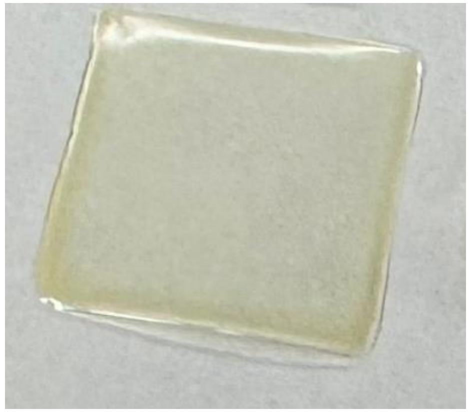
[BPTA], **[BPTA/BPPA = 7:3], [BPTA:BPPA = 7:3]**, [**BPPA**]

- A lower-MW **mTPPA (22 kDa)** was therefore evaluated

mTPPA Blends Underperform: Phosphonic Acid Acidity is the Limiting Factor

- BPSA/mTPPA showed lower conductivity than BPSA under all tested conditions



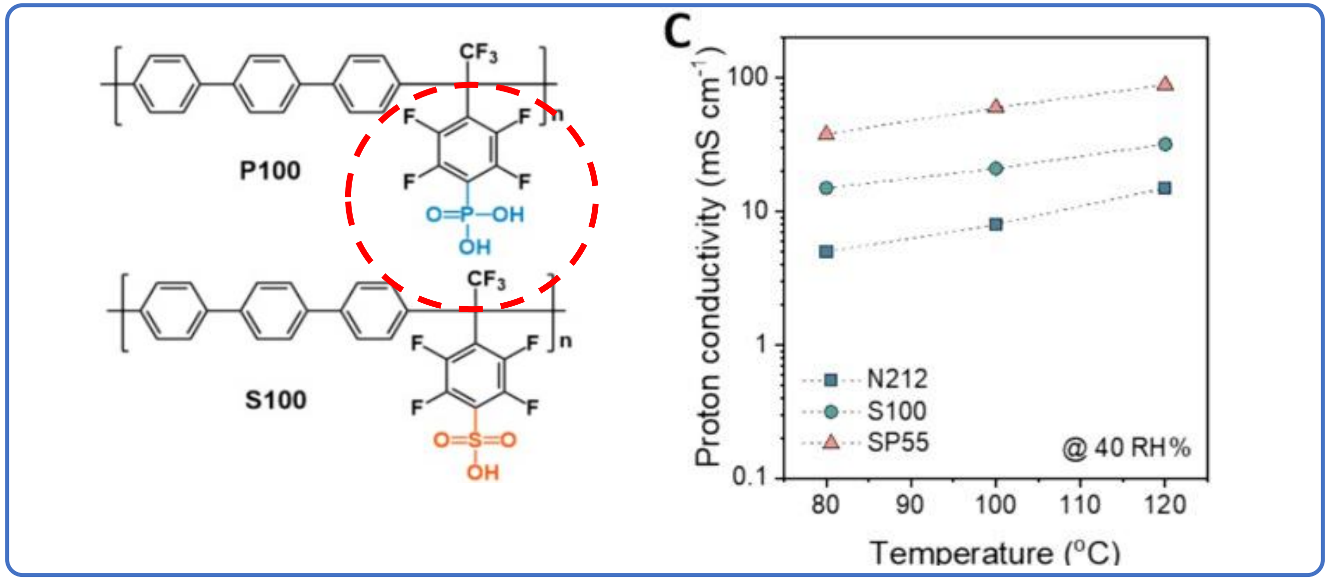


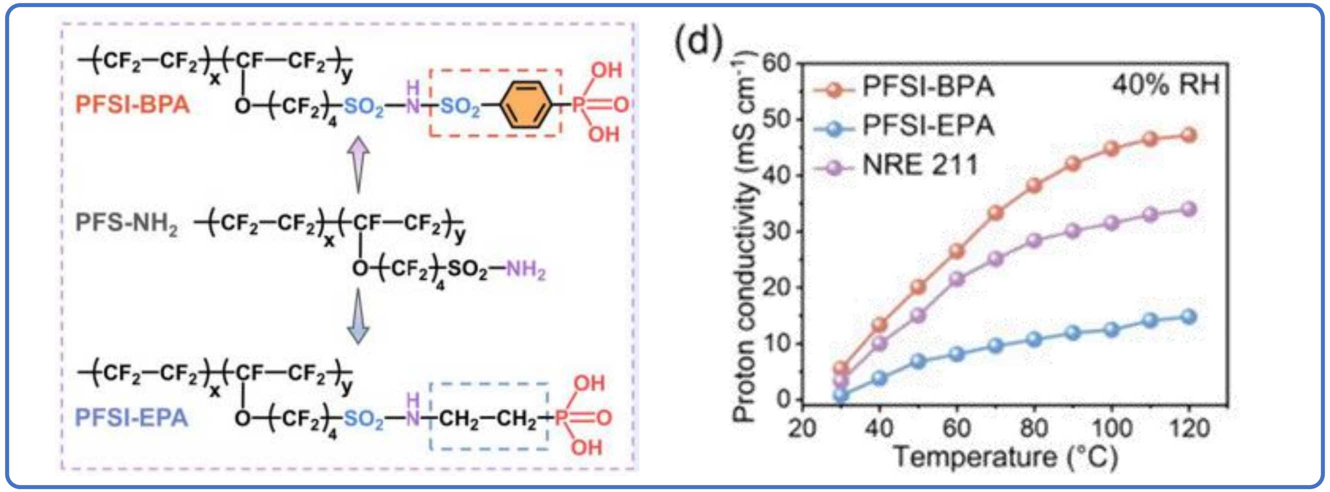
BPTA/mTPPA: 7:3

Condition		BPSA	BPSA/mTPPA 7:3	Nafion
24 °C	in water	84 ± 5	33 ± 8	
80 °C	in water	161	56	
80 °C	90% RH	72	37	114 ± 0
80 °C	40% RH	5	5	16 ± 0
120 °C	40% RH	5	0.059 (after 21 hours)	31 ±0

Root Cause: Weak Phosphonic Acid Acidity

- It seems that electron withdrawing moiety to phosphonate is needed

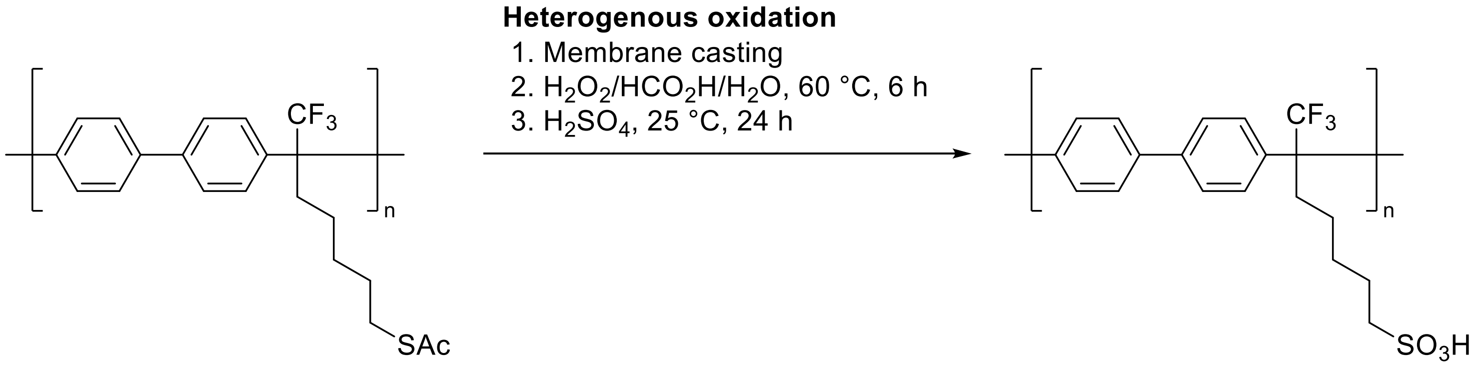


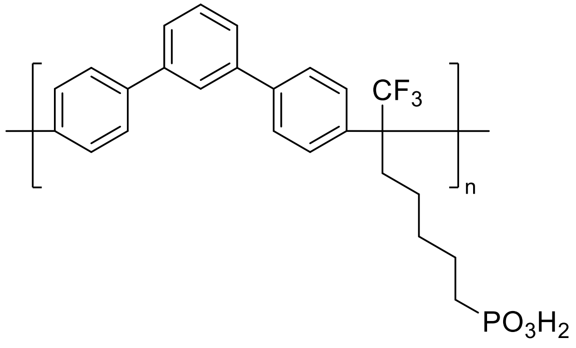


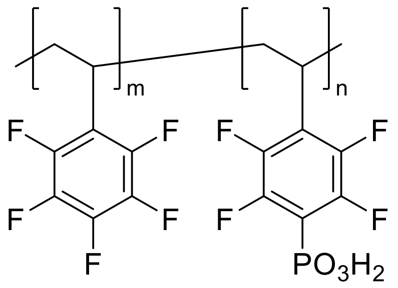
Chemical Engineering Journal **2025**, *522*, 167336; *ACS Nano* **2025**, *19*, 40536

Solution: Electron-Withdrawing Phosphonic Acid Improves Hot/Dry Conductivity

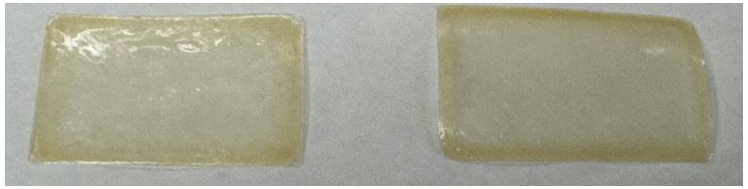
- BPSA/PWN blends showed the highest conductivity
- This identifies PWN as the more promising route for hot/dry proton transport.







BPTA 49 kDa **BPSA** **mTPPA** **PWN**



BPTA blended with
(L) PWN 43 and (R) PWN61 in 7:3

Condition		BPSA	BPSA/mTPPA 7:3	Nafion	BPSA/PWN43 7:3	BPSA/PWN61 7:3
80 °C	90% RH	72	37	114 ± 0	115	47 ± 2.8
80 °C	40% RH	5	5	16 ± 0	35	1.9 ± 0.21
120 °C	40% RH	5	0.059 (after 21 hours)	31 ±0	5	2.1 ± 0.1

Future Directions

1. Synthesizing new polymers with stronger electron-withdrawing groups

2. Optimizing phosphonated polymer ratio and molecular weight

3. Understanding morphology using SAXS to guide further design

Goal: polymer membranes that perform reliably at high temperature and low humidity.

Methodological Takeaways

Multimodal Characterization

NMR · Raman
UV–Vis · EIS

Structure-Property Thinking

From catalyst design to membrane optimization

Interdisciplinary Collaboration

Inorganic chemistry, polymer synthesis, electrochemistry

Independent Problem-Solving

Mechanistic probing through targeted molecular design

Thank You So Much for Your Attention!

Chuan Pin (Pin) Chen

yellowchen777@hotmail.com | Postdoctoral Appointee, Argonne National Laboratory

Sulfonation of BPBr